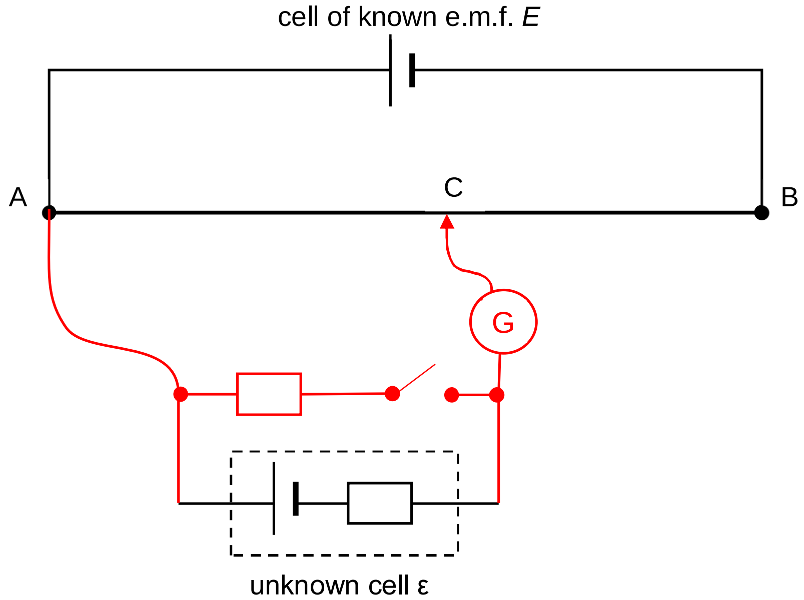
for noise waves of <u>longer periods</u>; correspond to <u>lower frequencies</u> A1

5(a)



- Correct drawing with switch in the correct position. B1

Turn off the switch. Find balance length when galvanometer shows null reading. B1
Find ε using $\varepsilon = V_{AC} = kL_{AC}$ and $V_{AB} = E = kL_{AB}$

Turn on the switch.
Find the new balance length, L_{AC}' B1

To find r:
Compare $V_{AC}' = kL_{AC}'$ and $V_{AB} = E = kL_{AB}$
$V_{AC}' = kL_{AC}' = (R/R+r).\ \varepsilon$
Solve r.

Question		Marks
5(b)(i)	For single strand: $R = \dfrac{\rho L}{A}$ $= \dfrac{(4.9 \times 10^{-7})(1)}{\pi \left(\dfrac{0.15 \times 10^{-3}}{2} \right)^2}$ (= 27.7 Ω)	M1
	For 7 strands: $\dfrac{1}{R_{eff}} = 7\left(\dfrac{1}{R}\right)$ $R_{eff} = \dfrac{R}{7}$ $= \dfrac{(4.9 \times 10^{-7})(1)}{\pi (7) \left(\dfrac{0.15 \times 10^{-3}}{2} \right)^2}$ = 3.96 Ω = 4.0 Ω	A1
5(b)(ii)	Potential difference across the 7 Ω resistor = $\frac{7}{14+7} \times 12 = 4\ V$	B1
	By potential divider rule, potential difference across 0.9 m potentiometer $= \frac{0.9 \times 3.96}{3.96+10} \times 12 = 3.06\ V$	B1
	Since the potential difference across the potentiometer is less than the potential difference across the 7 Ω resistor, balance length cannot be achieved.	B1
6(a)	Work function is the minimum amount of energy required for an electron to	B1